EX.01: Designing a responsive layout for a societal application

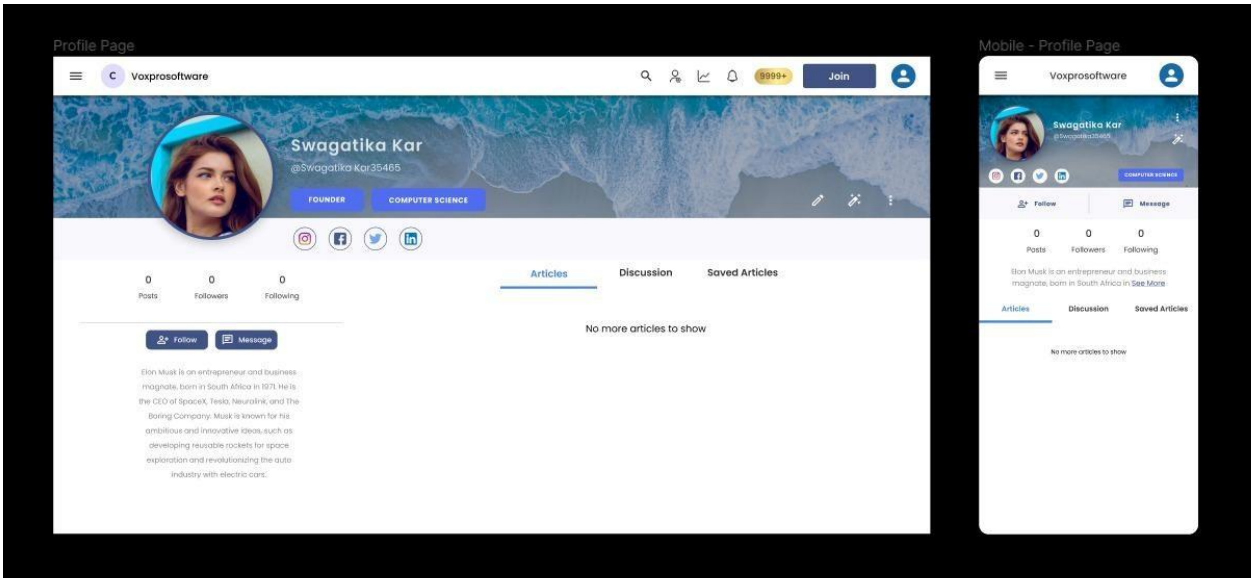
Aim:

To design a responsive layout for a societal application.

Algorithm/Procedure:

1.Creating a responsive website in Figma:
2.Define your website's purpose and audience.
3.Set breakpoints for different screens.
4.Design desktop version.
5.Use grids for layouts.
6.Design for tablet and mobile.
7.Create responsive components.
8.Apply constraints to elements.
9.Use relative font sizes.
10.Optimize images.
11.Test and prototype.
12.Use grid/ flex box layouts.
13.Design responsive navigation.
14.Consider touch interactions.
15.Test on various devices.
16.Iterate based on feedback.
17.Document choices.
18.Share with developers.
19.Maintain and update design.

Output:



Result:

Thus designing of responsive layout for a societal application has been performed successfully.

EX.02: Exploring various UI Interaction Patterns

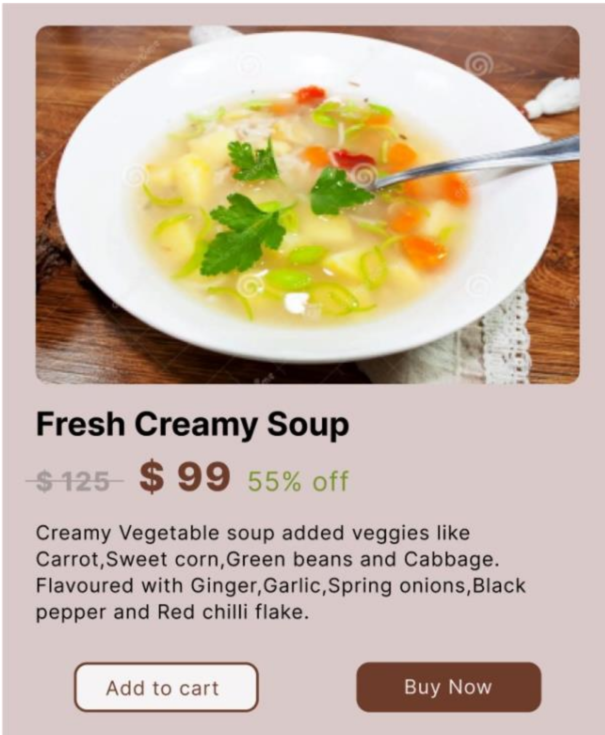
Aim:

To explore various UI interaction patterns.

Algorithm/Procedure:

1.Set objectives and understand user needs.
2.Research and gather design inspiration.
3.Create wireframes for layout and structure.
4.Utilize Figma components and styles.
5.Prototype interactions using Figma's features.
6.Test your design with users for feedback.
7.Iterate and refine based on feedback.
8.Document your design decisions.

Output:



Result:

Thus various UI interaction patterns have been explored successfully.

Ex.03: Developing an Interface with Proper UI Style Guides

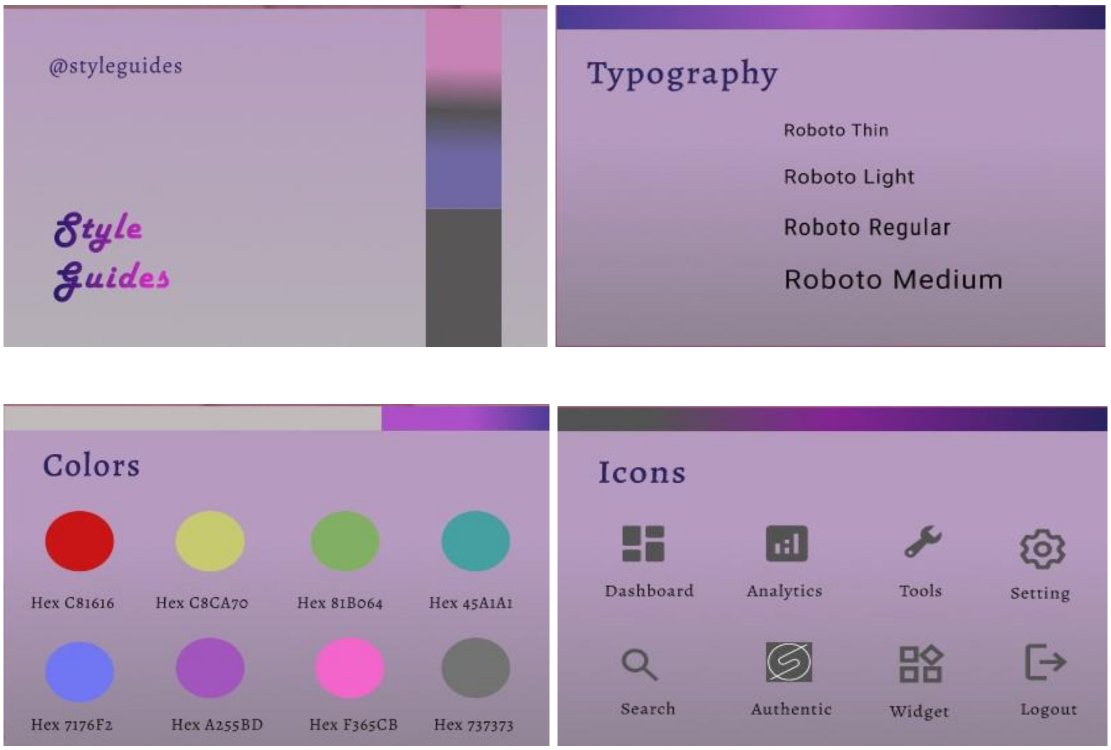
Aim:

To develop an Interface with Proper UI Style Guides.

Algorithm/Procedure:

1. Define Goals and Audience: Understand the project's purpose and target users.
2. Research and Inspiration: Gather industry insights and design inspiration.
3. Create a Figma Project: Start a new Figma project.
4. Workspace Setup: Organize Figma files and create sections for style guides and components.
5. Brand Guidelines: Set color, typography, and brand-related guidelines.
6. UI Components: Create component libraries for buttons, forms, icons, and navigation.
7. Typography: Define font styles, sizes, and spacing.
8. Iconography: Design and organize icons as components.
9. Color System: Document primary, secondary, background, and text colors.
10. Grids and Layouts: Establish grid systems for different devices. 11. Accessibility Guidelines: Ensure WCAG-compliant contrast and text sizing.
12. UI Elements: Document usage guidelines for various UI elements.
13. Wireframes: Create wireframes and user flows for screen layout and interaction
14. Visual Design: Develop screens based on wireframes using components and styles.
15. Interactive Prototyping: Build interactive prototypes for user testing and feedback. Downloaded

Output:



Result: Thus an Interface with Proper UI Style Guides has been developed successfully.

Ex.04: Developing wire flow diagram for application using open-source software

Aim:

To develop Wire flow diagram for application using open-source software.

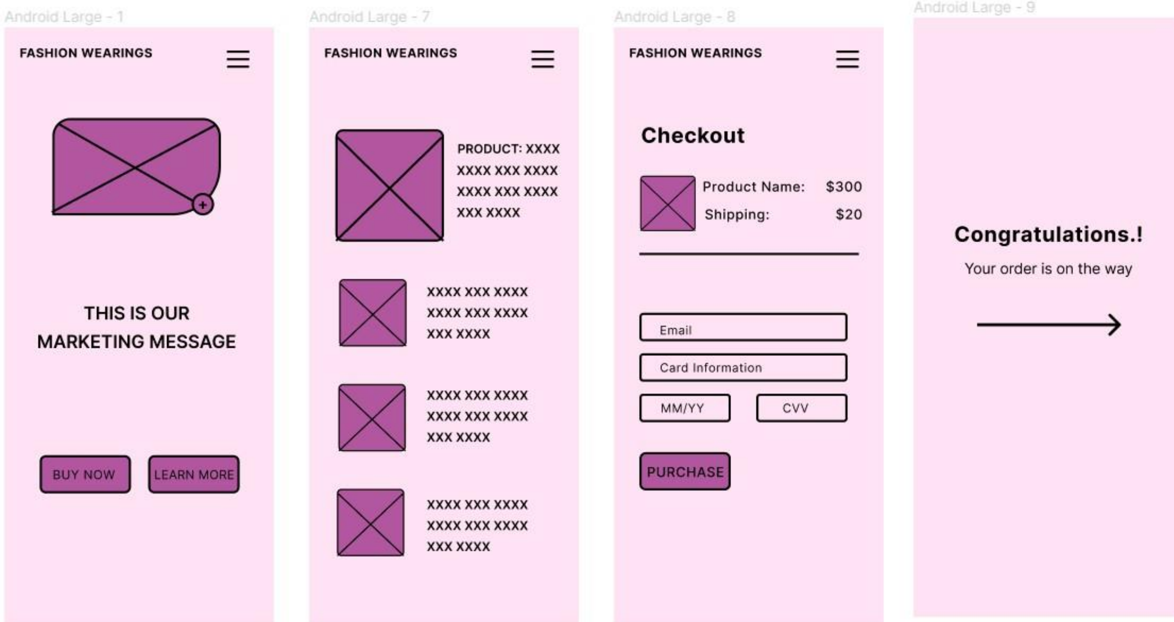
Algorithm/Procedure:

1. Define Purpose and Goals: Determine the diagram's purpose and goals, focusing on user flows, navigation, and interactions.
2. Identify User Personas: If applicable, specify user personas for a user-centric approach.
3. Gather Requirements: Collect project information, including existing designs and functionality requirements.
4. Select Software: Choose open-source design software,

a. such as Figma, for wire flow creation.

5. Create a Project: Begin a new project in your chosen software and set up the canvas to match your project's needs.
6. Wireframe Screens: Develop wireframes for each application screen, focusing on visual structure.
7. Define Interactions: Add interaction notes or links to illustrate navigation and user interactions.
8. Create User Flows: Connect wireframes to illustrate user journeys, navigation paths, and interactions.
9. Add Annotations: Include descriptions to clarify elements and interactions in each wireframe.
10. Collaborate and Share: Utilize collaboration features to gather feedback from team members and stakeholders
11. Iterate and Refine: Revise the wire flow diagram based on - feedback, ensuring alignment with project goals.
12. Finalize and Export: Clean up the wire flow diagram and export it to a suitable format for sharing and documentation.
13. Document the Wire flow: Create a reference guide to explain the wire Flow’s purpose and key notes for stakeholders and developers.

Output:



Result:

Thus Wire flow diagram for application using open-source software has been developed successfully.

Ex.05: Hands on Design Thinking Process for a new product

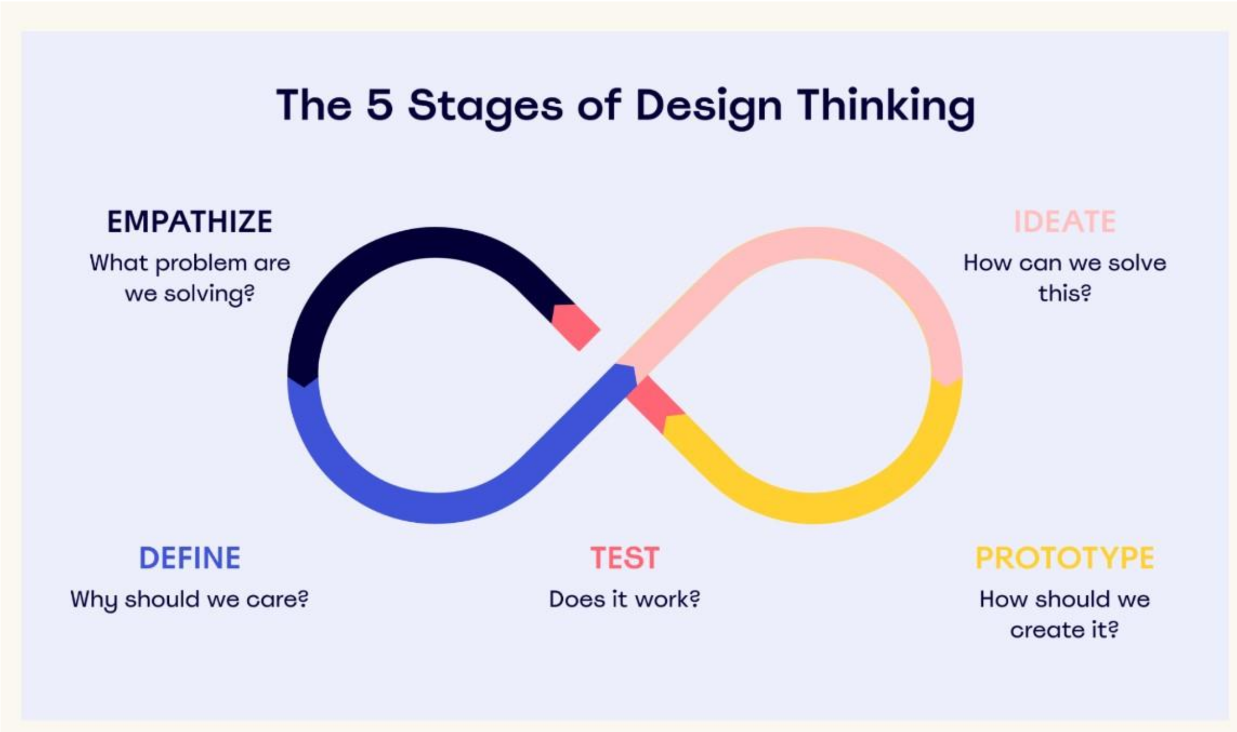
Aim:

To apply the design thinking process for a new product.

Algorithm/Procedure:

Empathize: Begin by conducting user research and interviews to gain insights into potential user needs and pain points related to smartphone usage.
Define: Analyze the gathered information to define a clear and specific problem statement. For example, "Users need a more efficient way to track their daily fitness activities."
Ideate: Organize brainstorming sessions with a diverse team to generate a wide range of creative solutions. Encourage free thinking and open collaboration.
Prototype: Create a low-fidelity prototype of the smartphone app. This can be a paper sketch or a digital wireframe that represents the app's basic functionality.
Test: Conduct user testing sessions with a small group of potential users. Observe how they interact with the prototype and gather feedback.
Iterate: Based on user feedback, refine the prototype and make necessary improvements to address user concerns or suggestions.
Prototype (Again): Create a more advanced prototype, closer to the final product. It should incorporate the changes and improvements identified during the initial testing phase.
Test (Again): Conduct another round of user testing, this time with a larger group of users. Gather data on usability, functionality, and overall user experience.
Refine: Analyze the results of the second testing phase and make further refinements to the app design and functionality.
Implement: Develop the final version of the smartphone app, incorporating all the changes and improvements identified during the design thinking process.
Test (Final Testing): Conduct thorough testing of the fully developed app to ensure it's bug-free and ready for launch.
Launch: Release the app to the target market, accompanied by marketing and promotion efforts.

Output :



Result:

Thus the design thinking process for new product has been studied.

Ex.06: Brainstorming feature for proposed product

Aim:

The aim of this process is to generate innovative and practical feature ideas for a proposed product, ensuring that the final product meets user needs, addresses pain points, and has a competitive edge in the market.

Algorithm/Procedure:

Understand the Product Concept:

Begin by thoroughly understanding the proposed product's concept, its target audience, and its unique selling points.

Gather a Diverse Team:

Assemble a cross-functional team with members from various departments (e.g., product development, marketing, design) to bring different perspectives to the brainstorming session. **Set Clear Objectives:**

Define clear objectives for the brainstorming session. What problems should the new features solve? What goals should they achieve?

Warm-Up and Icebreaker:

Start the session with a warm-up or icebreaker activity to encourage creative thinking and open communication within the team.

Idea Generation:

Allow team members to freely brainstorm feature ideas. Encourage a "no idea is a bad idea" mindset. Use techniques like mind mapping, brainstorming software, or post-it notes on a whiteboard to record ideas.

Categorize and Prioritize:

Group similar ideas together, and prioritize them based on factors like feasibility, potential impact, and alignment with the product concept.

SWOT Analysis:

Conduct a SWOT (Strengths, Weaknesses, Opportunities, Threats) analysis for each feature idea to evaluate its potential in the market.

Feasibility Assessment:

Assess the technical, financial, and resource feasibility of implementing the proposed features.

Market Research:

Conduct market research to identify user preferences and gather insights that can inform feature development. **Prototype and User Testing:**

Create prototypes or mockups of the proposed features and conduct user testing to gather feedback and refine the ideas.

Cost-Benefit Analysis:

Evaluate the expected cost of development against the projected benefits, such as increased user engagement, retention, or revenue.

Risk Assessment:

Identify potential risks associated with each feature and develop mitigation strategies.

Finalize Feature Set:

Based on the assessment, finalize the set of features to be included in the product.

Ensure they align with the product's vision and goals.

Documentation:

Document the chosen features, their objectives, and the rationale behind their selection.

This document will guide the development team.

Iterate as Needed:

Keep an open line of communication for ongoing feature refinements and iterations, especially as more data and insights become available.

Example:

Suppose a software company is developing a new mobile messaging app. During the brainstorming session, the team generates a wide range of feature ideas, including:

End-to-End Encryption: To ensure user privacy and data security.

Message Scheduling: Allowing users to schedule messages to be sent at a specific time.

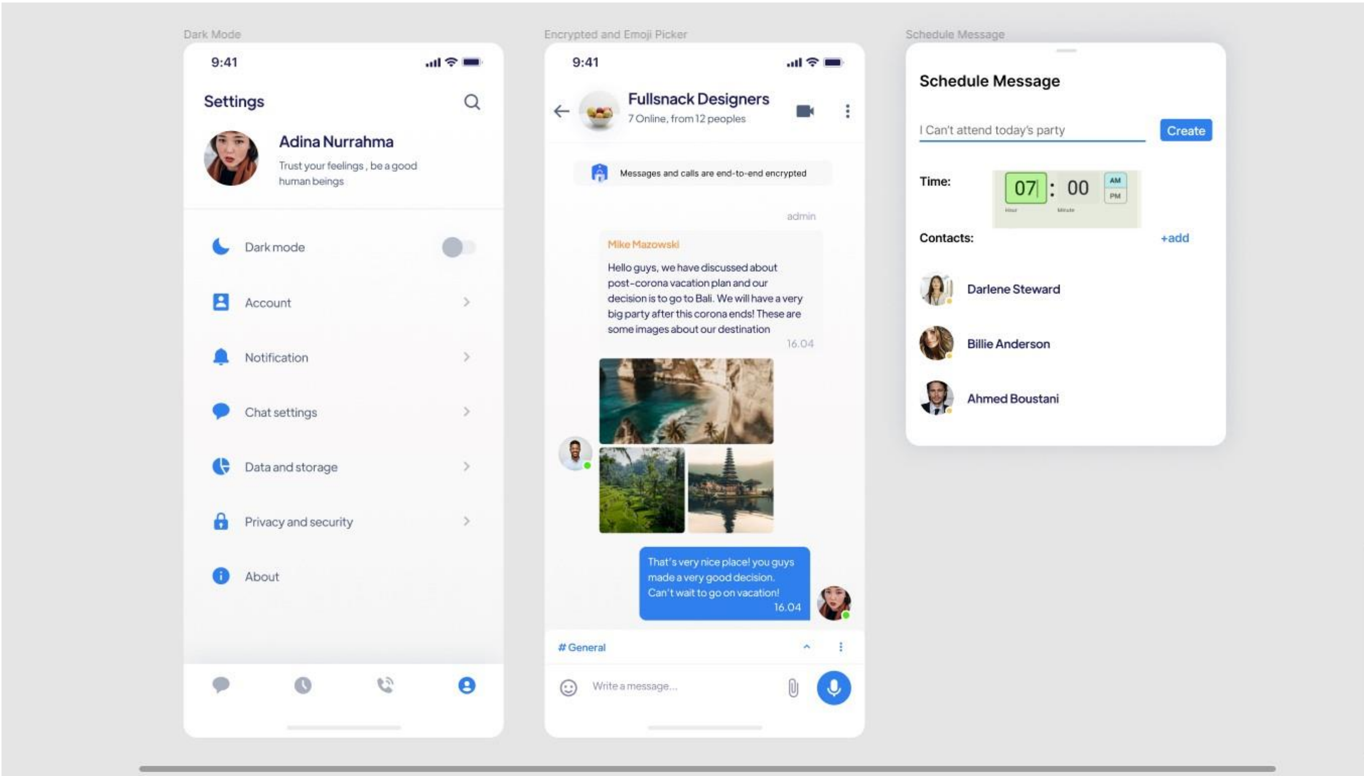
Reaction Emojis: A feature that lets users react to messages with emojis for more expressive communication.

Dark Mode: A night-friendly theme for the app.

Polls and Surveys: Integration of polls and surveys within the chat for easy decisionmaking.

Auto-Translate: Real-time language translation for international communication.

Output:



Result:

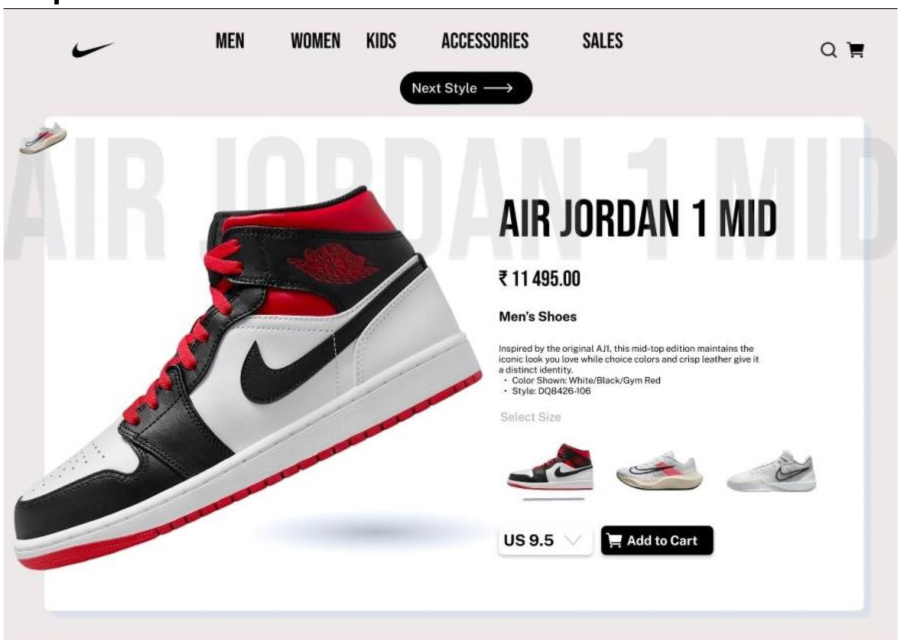
Thus brainstorming feature for proposed product has been applied and executed successfully.

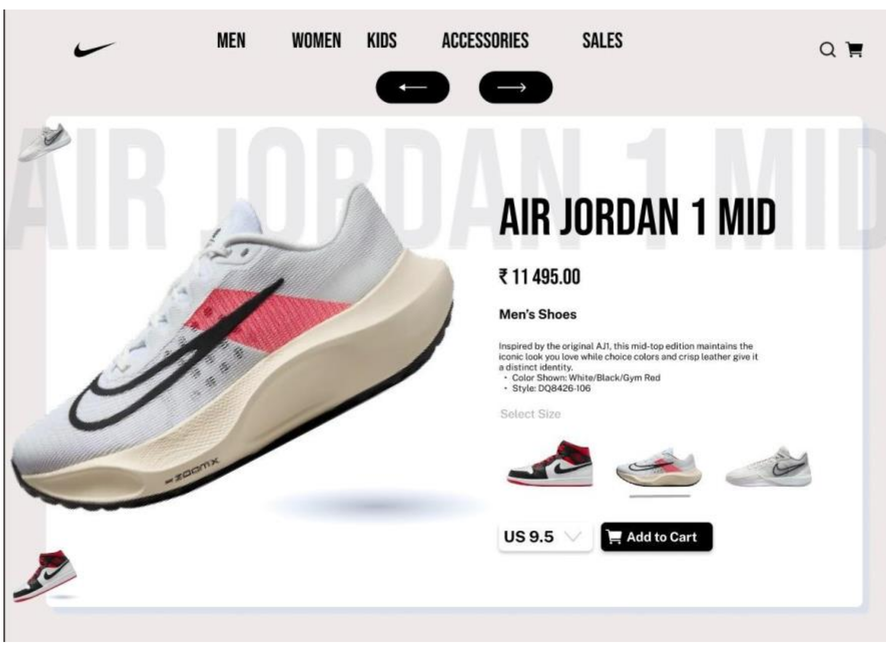
Ex.07: Defining the Look and Feel of the new Project

Aim:
The aim is to establish the visual design and user experience for a new project, ensuring it aligns with the project's goals and provides an appealing, intuitive, and cohesive interface for users.

Algorithm/Procedure:
1.Project Goal Assessment:
Understand the project's objectives, target audience, and scope. This sets the foundation for design decisions.
2.Research and Inspiration:
Gather inspiration from existing designs and industry trends. Create mood boards or design boards to collect visual references.
3.Define Design Principles:y, accessibility, and branding.
4.Wireframing and Prototyping:
Create wireframes or low-fidelity prototypes to plan the layout and structure of the user interface. Use tools like Figma, Sketch, or Adobe XD for digital projects.
5.Visual Design:
Develop a color palette, typography choices, and graphic elements (icons, images, logos) that reflect the project's identity. Create high-fidelity designs using design software.
6.User Interaction Design: Define user interactions and behaviors, including animations, transitions, and microinteractions.
Ensure a smooth and intuitive user experience.
7.Responsive Design:
Adapt the design to various screen sizes and devices, focusing on mobile responsiveness.
8.Accessibility and Usability Testing:
Evaluate the design for accessibility, ensuring it's usable by individuals with disabilities. Conduct usability testing with potential users to gather feedback.
9.Iteration and Feedback:
Refine the design based on feedback from users and stakeholders.
10.Documentation:
Create design documentation that includes guidelines for developers to implement design.
11.Development Integration:

Output:





Result:

Thus the Look and Feel of the new Project has been defined successfully

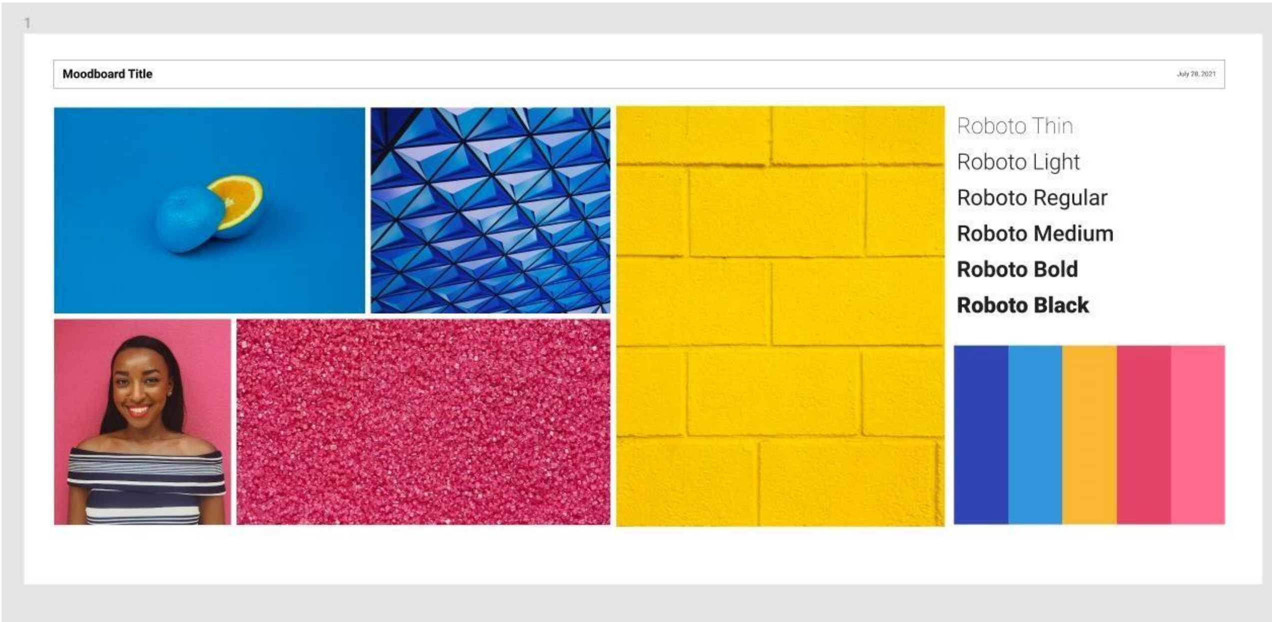
EX.8:Create a sample pattern library for product(mood board, font and colors based on principles)

Aim:
To develop a pattern library for product that involves creating mood board,fonts and colors based on principles.

Algorithm:
1.**Establish Product Vision**:Define the overall vision and target audience for your product. This sets the tone for the mood board, font choices, and design principles.

2.**Create a Mood Board**:Compile images, textures, and color palettes that reflect the desired mood or theme.Include a variety of imagery (abstract, realistic, etc.) to inspire creativity and direction.
3.**Select a Color Palette**:Define a set of primary colors that align with the product's theme.Add secondary colors for accent and contrast.Include neutral tones for balance.
4.**Choose Typography**:Select a font family for headings and another for body text. Ensure readability and consistency.Incorporate font weights for different levels of emphasis (bold, regular, light, etc.).
5.**Develop Layout Templates**:Create templates for common layouts (e.g., homepage, product page, contact page).Use a grid system to maintain consistency in alignment and spacing.
6.**Define Design Principles**:Establish principles for visual hierarchy, consistency, and responsiveness.Include guidelines for accessibility (e.g., color contrast, screen reader compatibility).
7.**Design Components and Controls**:Develop consistent designs for buttons, forms, and other interactive elements.Ensure these components follow the established color palette and typography.
8.**Create Icons and Graphics:**Develop a set of icons that align with the product's theme and are easily recognizable.Use consistent styles and proportions for icons.
9.**Document the Pattern Library**:Create a guide that documents all components, design principles, and examples.Include code snippets or style sheets for developers to implement the pattern library.
10.**Implement a Review and Feedback Process**:Establish a process to review new design elements for consistency with the pattern library.

OUTPUT:



RESULT:

This implementation of sample pattern library has been implemented successfully.

Ex.09: Identify a customer problem to solve

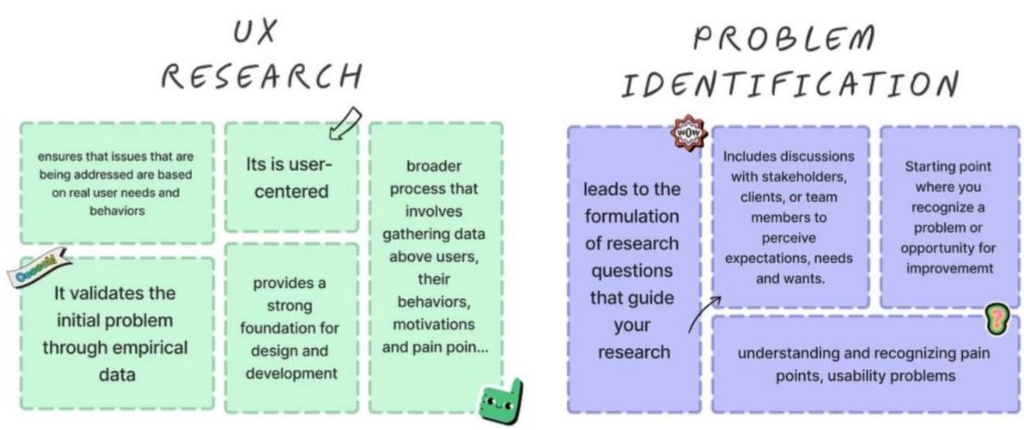
Aim:
The aim of this experiment is to identify a customer problem to solve effectively, which is crucial for product development, customer satisfaction, and business success.

Algorithm/Procedure:

1. Customer Segmentation:Begin by segmenting your customer base into different groups based on demographics, behavior, or other relevant criteria.
2.Data Collection: Gather data from these customer segments through surveys, interviews, feedback forms, and analytics tools. You can also utilize data from your customer support system, website, or app analytics.
3. Problem Identification Metrics: Define key metrics and indicators to identify customer problems. Examples include high bounce rates on a specific webpage, low customer satisfaction scores, or a surge in support tickets related to a specific issue.
4.Data Analysis: Analyze the collected data to identify patterns, trends, and common issues reported by customers. Data analysis tools and techniques, such as data mining or sentiment analysis, can be useful.
5. Prioritization: Prioritize the identified problems based on their impact on customers and your business. You can use techniques like the Moscow method (Must- haves, Should-haves, Could-haves, Won't-haves) to prioritize.
6.Root Cause Analysis: Conduct a root cause analysis for each identified problem. Understand why these issues are occurring by delving into the underlying causes. **7.Solution Ideation:** Brainstorm potential solutions for the identified problems. Encourage cross-functional teams to contribute ideas and consider how these solutions align with your business goals.
8.Experiment Design: Design controlled experiments or A/B tests to validate the proposed solutions. Ensure you have a clear hypothesis and success criteria for each experiment.

9.Implementation: Implement the proposed solutions on a small scale to observe their impact. This might involve website changes, process adjustments, or feature additions.
10.Data Collection Post-Implementation: Continue to collect data after implementing the solutions to assess their effectiveness. Monitor key metrics to see if they improve.
11.Analysis and Validation: Analyze the post-implementation data to validate whether the proposed solutions have effectively addressed the customer problem. Make data- driven decisions.
12.Feedback and Iteration: Collect feedback from customers regarding the changes and iterate on the solutions based on their input. Continuous improvement is key.

Output:



Result: Thus a customer problem was identified and understood successfully.

Ex.10: Sketch, design with popular tool and build a prototype and perform usability testing and identify improvements

Aim:

The aim of this experiment is to design a user-friendly mobile app for task management, create a prototype using a popular design tool, perform usability testing, and identify improvements to enhance the user experience.

Algorithm/Procedure:

1.Define Objectives and User Persona: Define the objectives of the task management app. Create a user persona to represent the target audience.
2.Sketch and Wireframe: Start with sketching the basic layout and functionality of the app on paper or digitally. Create low-fidelity wireframes to visualize the app's structure and layout. **3.Design with a Popular Tool:** Choose a popular design tool such as Adobe XD, Sketch, Figma, or In Vision. Create high fidelity designs with attention to visual elements, typography, and color schemes. Implement the user interface (UI) based on best practices and your user persona's preferences.
4. Prototype Creation: Use the design tool to create interactive prototypes with clickable elements and transitions. Ensure that the prototype represents the app's core functionalities. **5.Recruit Participants for Usability Testing:** Identify potential users or participants who match the user persona. Prepare a usability testing plan, including tasks to be performed within the prototype.

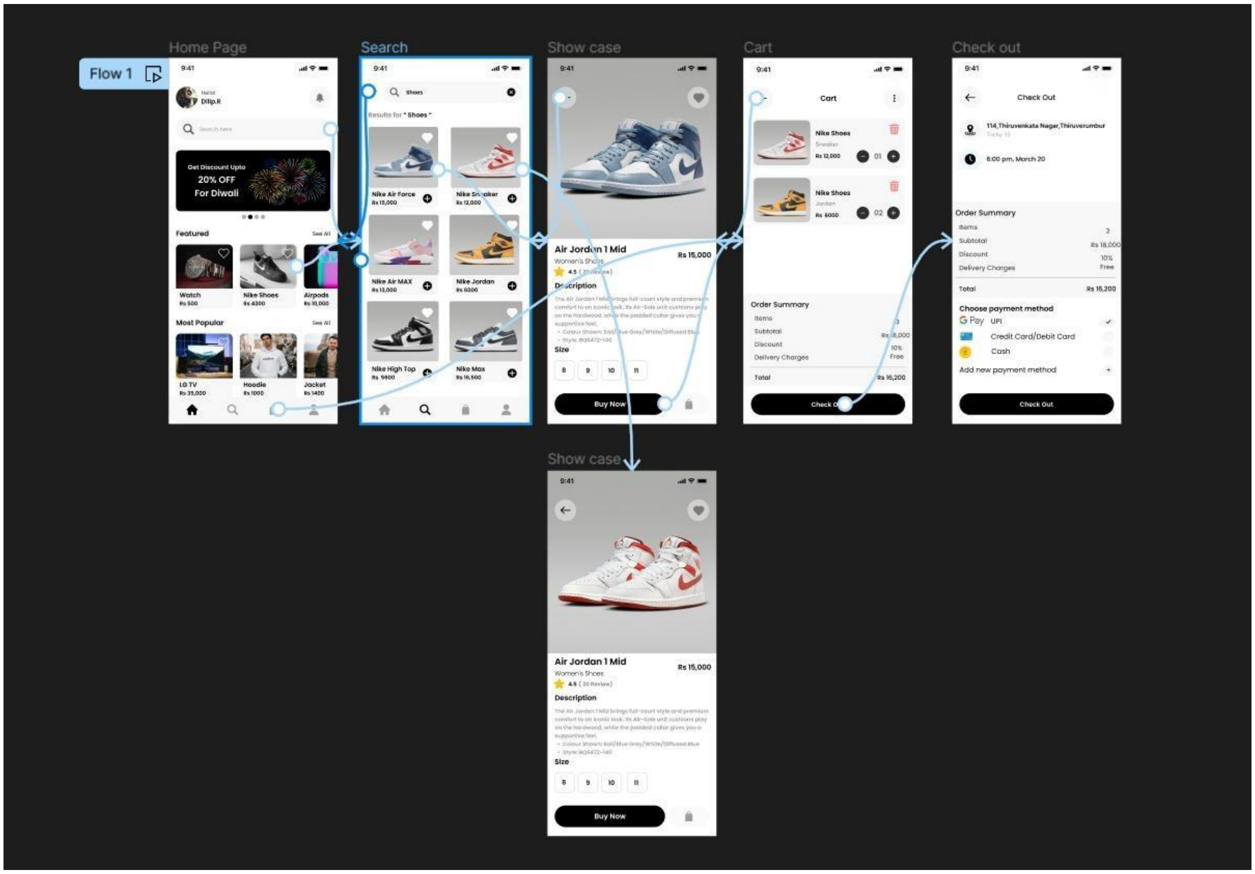
6. **Usability Testing:** Conduct usability testing sessions with participants. The participants are asked to perform specific tasks within the prototype. Observe and record their interactions and gather feedback on their experience.
7. **Analyze and Identify Improvements:** Analyze the usability testing data to identify pain points and areas of improvement.Look for common patterns and issues encountered by users.
8. **Iterate on the Design:** Implement the necessary design improvements based on the feedback received.Make changes to the prototype to address identified issues.

9. **Second Round of Usability Testing:** Conduct a second round of usability testing with new or the same participants to evaluate the impact of the design improvements

10. **Finalize the Prototype:** Make any final adjustments based on the results of the second usability testing round.

11. **Document Findings and Recommendations:** Document the findings from both rounds of usability testing. Provide clear recommendations for further improvements or development. **12.Conclusion:** Conclude the experiment by summarizing the improvements made to the prototype and how they enhance the user experience.

Output:



Result:

Thus, Sketching, building a prototype, performing usability testing and identifying improvements has been executed successfully.